Problem · Unit 3 · 10 points · 60 min · Standard exam level

P12 — Block on an accelerating turntable: the friction cone, the instant of slip and the direction in which the block starts to slide

A horizontal turntable rotates about the fixed vertical axis through its centre O. It starts from rest at $t = 0$ with constant angular acceleration $\beta > 0$, so its angular rate is $\Omega(t) = \beta t$ (counter-clockwise seen from above). A small block P of mass m rests on the table at the distance r_0 from the axis; at $t = 0$ it lies on the fixed x axis. The contact is rough, with static Coulomb coefficient μ and kinetic coefficient $\mu_k \le \mu$; gravity is $-g\,\mathbf{k}$.

Use the inertial frame $S_0 = \{O; \mathbf{i}, \mathbf{j}, \mathbf{k}\}$ ($\mathbf{k}$ vertical, upward), the turntable frame S_1 (rotating with the table), the angle φ of $\overrightarrow{OP}$ from the x axis and the unit vectors $\mathbf{e}_r = \cos\varphi\,\mathbf{i} + \sin\varphi\,\mathbf{j}$, $\mathbf{e}_\varphi = \cos\varphi\,\mathbf{j} - \sin\varphi\,\mathbf{i}$.

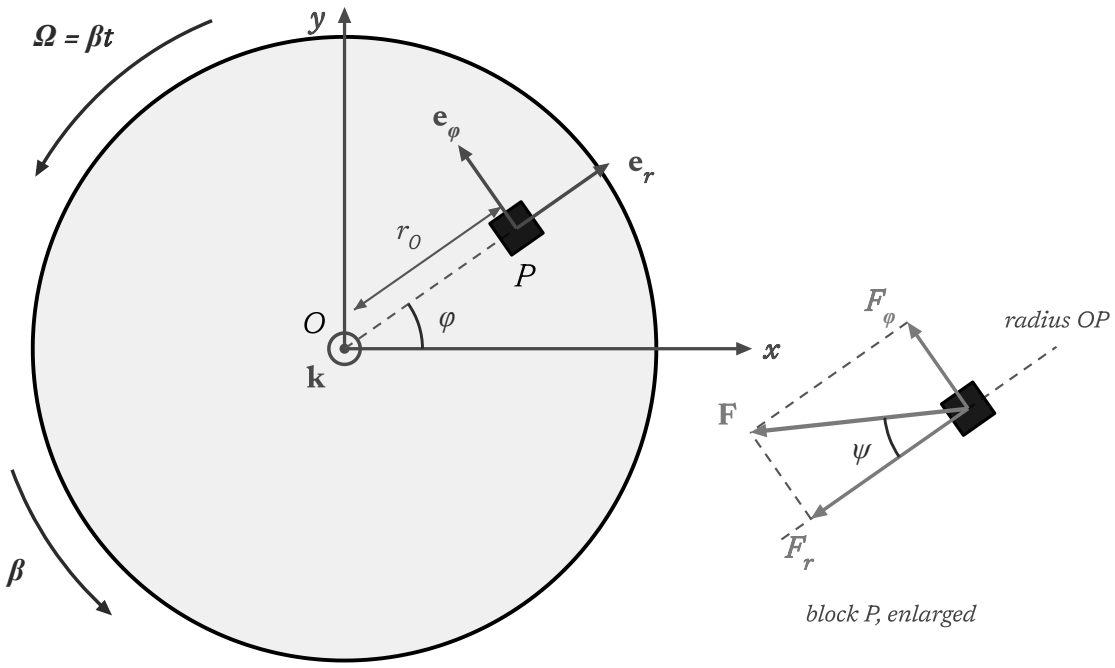


Figure 1. Turntable seen from above: fixed axes x, y and $\mathbf{k}$ (out of the page, upwards) at the axis O, rate $\Omega = \beta t$ and angular acceleration β (counter-clockwise), block P at distance r_0; while it sticks, the friction $\mathbf{F}$ has a component $F_r < 0$ towards O and a component $F_\varphi > 0$ along $\mathbf{e}_\varphi$.

Setup

- Data: m, g, μ, μ_k, r_0, β. Numbers: $\mu = 0.4$, $\mu_k = 0.3$, $r_0 = 0.5$ m, $\beta = 0.2$ rad s^{-2}, $g = 9.81$ m s^{-2}.
- Hypotheses: point particle (the block does not tip), rigid horizontal table, Coulomb friction, no air drag.
- In (e) and (f) the friction force on P is written $\mathbf{F} = F_r\,\mathbf{e}_r + F_\varphi\,\mathbf{e}_\varphi$ and ψ is its angle from the inward radius $-\mathbf{e}_r$, positive towards $+\mathbf{e}_\varphi$ (figure).

Part (a) · 1 point

Degrees of freedom and frames

How many degrees of freedom has P while it sticks to the table, and while it slides on it? Give the angular velocity and the angular acceleration of S_1 relative to S_0.

Step 1. Count

A free particle has 3 degrees of freedom. While it touches the table, $z = 0$ (bilateral as long as $N \geq 0$): 2 remain, the polar coordinates (r, φ). While it does not slip, it moves with the table: $r = r_0$ and φ is the angle turned by the table, $\varphi = \int_0^t \beta\tau\,d\tau = \left[\tfrac{1}{2}\beta\tau^2\right]_0^t = \tfrac{1}{2}\beta t^2$. Nothing is left free: 0 degrees of freedom, and the friction force is an **unknown reaction** that adjusts to keep the block in place, as long as it can (Notes §6.6).

Step 2. Frames

S_1 turns about $\mathbf{k}$ with $\boldsymbol{\omega}_{10} = \Omega\,\mathbf{k} = \beta t\,\mathbf{k}$; its angular acceleration $\dot{\boldsymbol{\omega}}_{10} = \beta\,\mathbf{k}$ is not zero, so in S_1 the block feels an Euler force as well as the centrifugal and Coriolis forces.

KEY RESULTS

- Sliding: 2 degrees of freedom (r, φ) on the plane $z = 0$. Sticking: 0 degrees of freedom, $r = r_0$, $\varphi = \tfrac{1}{2}\beta t^2$ prescribed.
- $\boldsymbol{\omega}_{10} = \beta t\,\mathbf{k}$, $\dot{\boldsymbol{\omega}}_{10} = \beta\,\mathbf{k}$.

Part (b) · 1.5 points

Kinematics while the block sticks

Compute the velocity and acceleration of P in S_0 in the basis $\{\mathbf{e}_r, \mathbf{e}_\varphi\}$.

Step 1. Change of basis and unit vectors

The polar basis $\{\mathbf{e}_r, \mathbf{e}_\varphi, \mathbf{k}\}$ turns with the table; its relation to the inertial basis is

$$\begin{pmatrix} \mathbf{e}_r \\ \mathbf{e}_\varphi \\ \mathbf{k} \end{pmatrix} = [R] \begin{pmatrix} \mathbf{i} \\ \mathbf{j} \\ \mathbf{k} \end{pmatrix}, \qquad [R] = \begin{pmatrix} \cos\varphi & \sin\varphi & 0 \\ -\sin\varphi & \cos\varphi & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Checks: rows 1 and 2 have length $\cos^2\varphi + \sin^2\varphi = 1$ and $-\cos\varphi\sin\varphi + \sin\varphi\cos\varphi = 0$, row 3 is $\mathbf{k}$, so $[R][R]^T = I$; $\det[R] = \cos^2\varphi + \sin^2\varphi = +1$ (right-handed, $\mathbf{e}_r \times \mathbf{e}_\varphi = \mathbf{k}$). Then $\dot{\mathbf{e}}_r = \dot\varphi\,(-\sin\varphi\,\mathbf{i} + \cos\varphi\,\mathbf{j}) = \dot\varphi\,\mathbf{e}_\varphi$ and $\dot{\mathbf{e}}_\varphi = -\dot\varphi\,\mathbf{e}_r$ (chain rule through $\varphi(t)$).

Step 2. Velocity and acceleration

$\mathbf{r} = r_0\,\mathbf{e}_r$ with r_0 constant:

$$\mathbf{v} = r_0\dot\varphi\,\mathbf{e}_\varphi = r_0\Omega\,\mathbf{e}_\varphi$$

$$\mathbf{a} = r_0\ddot\varphi\,\mathbf{e}_\varphi + r_0\dot\varphi\,\dot{\mathbf{e}}_\varphi = r_0\beta\,\mathbf{e}_\varphi - r_0\Omega^2\,\mathbf{e}_r$$

The general polar formula $\mathbf{a} = (\ddot r - r\dot\varphi^2)\mathbf{e}_r + (r\ddot\varphi + 2\dot r\dot\varphi)\mathbf{e}_\varphi$ gives the same with $\dot r = \ddot r = 0$.

KEY RESULTS

- $\mathbf{v} = r_0\Omega\,\mathbf{e}_\varphi$.
- $\mathbf{a} = -r_0\Omega^2\,\mathbf{e}_r + r_0\beta\,\mathbf{e}_\varphi$: centripetal **and** tangential parts.

Part (c) · 2 points

Forces and the friction needed to stick

List the forces on P and find the normal reaction and the friction force needed for the block to move with the table. Check the result in S_1.

Step 1. Forces

Weight $-mg\,\mathbf{k}$; normal reaction $N\,\mathbf{k}$ ($N \geq 0$); friction $\mathbf{F} = F_r\,\mathbf{e}_r + F_\varphi\,\mathbf{e}_\varphi$ in the plane of contact, unknown while the block sticks.

Step 2. Newton in S_0

$m\mathbf{a} = -mg\,\mathbf{k} + N\,\mathbf{k} + \mathbf{F}$. Projecting:

$$\mathbf{k}:\ 0 = N - mg \Rightarrow N = mg; \qquad \mathbf{e}_r:\ -mr_0\Omega^2 = F_r; \qquad \mathbf{e}_\varphi:\ mr_0\beta = F_\varphi$$

Friction must pull **inward** (to bend the path) and **forward** (to speed the block up along with the table).

Step 3. Check in S_1 (Notes §4.9)

In S_1 the block is at rest, $\mathbf{v}_{rel} = \mathbf{0}$, so the Coriolis force vanishes. Centrifugal: $-m\boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r}) = -m\Omega^2\,\mathbf{k} \times (\mathbf{k} \times r_0\mathbf{e}_r) = -m\Omega^2\,\mathbf{k} \times r_0\mathbf{e}_\varphi = mr_0\Omega^2\,\mathbf{e}_r$. Euler: $-m\dot{\boldsymbol{\omega}} \times \mathbf{r} = -m\beta\,\mathbf{k} \times r_0\mathbf{e}_r = -mr_0\beta\,\mathbf{e}_\varphi$. Balance in the plane: $\mathbf{F} + mr_0\Omega^2\mathbf{e}_r - mr_0\beta\,\mathbf{e}_\varphi = \mathbf{0}$, the same F_r and F_φ.

KEY RESULTS

- $N = mg;\ F_r = -mr_0\Omega^2 = -mr_0\beta^2t^2;\ F_\varphi = mr_0\beta.$
- $|\mathbf{F}| = mr_0\sqrt{\Omega^4 + \beta^2}$, increasing in time.
- In S_1: $\mathbf{F} + mr_0\Omega^2\,\mathbf{e}_r - mr_0\beta\,\mathbf{e}_\varphi = \mathbf{0}$ (centrifugal and Euler forces, no Coriolis).

Part (d) · 2 points

When does the block slip?

Using the Coulomb condition, find the instant t^* at which the block starts to slide. What condition must β satisfy for the block to stick at all?

Step 1. The friction cone

Static friction holds while $|\mathbf{F}| \le \mu N$ (Notes §6.6). With (c):

$$m^2r_0^2\left(\Omega^4 + \beta^2\right) \le \mu^2m^2g^2 \iff \Omega^4 \le \left(\frac{\mu g}{r_0}\right)^2 - \beta^2$$

Step 2. Slip instant

The left side grows with t ($\Omega = \beta t$), so the inequality fails for the first time when it becomes an equality:

$$\Omega^* = \left[\left(\frac{\mu g}{r_0}\right)^2 - \beta^2\right]^{1/4}, \qquad t^* = \frac{\Omega^*}{\beta}$$

Step 3. Existence

A real $\Omega^* \geq 0$ needs $\beta \leq \mu g/r_0$. In the equality case $\beta = \mu g/r_0$ the formula gives $\Omega^* = 0, t^* = 0$: the friction is already at the limit of the cone at $t = 0$ and the block slides as soon as $\Omega > 0$, so it never sticks for a finite time. Sticking therefore requires the strict condition $\beta < \mu g/r_0$. If $\beta > \mu g/r_0$, already at $t = 0$ (where $\Omega = 0$) the tangential friction $mr_0\beta$ needed exceeds μmg: the table slides under the block from the start. Dropping β^2 in the cone (the usual shortcut $r_0\Omega^2 = \mu g$) hides exactly this case.

KEY RESULTS

- $r_0^2\left(\Omega^4 + \beta^2\right) \leq \mu^2 g^2$; slip when $\Omega^{*4} = \left(\dfrac{\mu g}{r_0}\right)^2 - \beta^2$.
- $t^* = \dfrac{\Omega^*}{\beta} = \dfrac{1}{\beta}\left[\left(\dfrac{\mu g}{r_0}\right)^2 - \beta^2\right]^{1/4}$; requires $\beta < \mu g/r_0$ (at $\beta = \mu g/r_0, t^* = 0$), otherwise it slips at $t = 0$.

Part (e) · 1.5 points

Numbers and the direction of the friction at the slip instant

For the data of the Setup compute Ω^*, t^*, the number of turns before slip and the speed of the block, and the angle ψ of the friction force at t^*.

Step 1. Instant of slip

$\mu g/r_0 = 0.4 \cdot 9.81/0.5 = 7.848\ \text{s}^{-2}$, so $(\mu g/r_0)^2 = 61.591\ \text{s}^{-4}$ and $\Omega^{*4} = 61.591 - 0.04 = 61.551\ \text{s}^{-4}$. Then $\Omega^{*2} = 7.8454\ \text{s}^{-2}$, $\Omega^* = 2.801\ \text{rad s}^{-1}$ and $t^* = 2.801/0.2 = 14.00$ s. Here $\beta^2/(\mu g/r_0)^2 = 6.5 \times 10^{-4}$: the tangential part barely changes t^*.

Step 2. Turns and speed

$\varphi^* = \frac{1}{2}\beta t^{*2} = \Omega^{*2}/(2\beta) = 7.8454/0.4 = 19.61$ rad, i.e. $19.61/2\pi = 3.12$ turns. The speed is $r_0\Omega^* = 1.400\ \text{m s}^{-1}$, along $\mathbf{e}_\varphi$.

Step 3. Direction of the friction

At t^*: $|F_r| = mr_0\Omega^{*2} = 3.923\,m$ N and $F_\varphi = mr_0\beta = 0.1\,m$ N, so $\tan\psi = \beta/\Omega^{*2} = 0.2/7.8454 = 0.0255$ and $\psi = 1.460^\circ$. Check: $\cos\psi = r_0\Omega^{*2}/(\mu g)$, and $|\mathbf{F}| = \mu mg = 3.924\,m$ N. The friction points to the axis, tilted 1.46° **forward**.

KEY RESULTS

- $\mu g/r_0 = 7.848\ \text{s}^{-2}, \Omega^{*4} = 61.551\ \text{s}^{-4}, \Omega^* = 2.801\ \text{rad s}^{-1}, t^* = 14.00$ s.
- $\varphi^* = \Omega^{*2}/2\beta = 19.61\ \text{rad} = 3.12$ turns; $|\mathbf{v}| = r_0\Omega^* = 1.400\ \text{m s}^{-1}$.
- $\tan\psi = \beta/\Omega^{*2} = 0.0255$, $\psi = 1.460^\circ$, inward and slightly forward.

Part (f) · 2 points

How does the block start to slide?

Just after t^* the friction is kinetic, of modulus $\mu_k mg$. Find the initial acceleration of P relative to the table, its modulus and its direction. Does the block leave along the radius? Describe its motion in S_0 at that instant.

Step 1. Inertia forces at t^*

In S_1, with $\mathbf{v}_{rel} = \mathbf{0}$, the inertia forces are those of (c): $\mathbf{F}_{in} = mr_0\Omega^{*2}\,\mathbf{e}_r - mr_0\beta\,\mathbf{e}_\varphi$. By (c) and (d), $\mathbf{F}_{in} = -\mathbf{F}_{needed}$ and $|\mathbf{F}_{in}| = \mu mg$. It points outward and **backward** (against the rotation), at the angle ψ from $+\mathbf{e}_r$.

Step 2. Direction of sliding

Once static friction fails, the block starts to move relative to the table along the unit vector $\hat{\mathbf{u}}$ of its relative acceleration, and kinetic friction opposes that motion: $\mathbf{F}_k = -\mu_k mg\,\hat{\mathbf{u}}$. In S_1, $m\mathbf{a}_{rel} = \mathbf{F}_{in} - \mu_k mg\,\hat{\mathbf{u}}$ must be parallel to $\hat{\mathbf{u}}$, which is possible only if $\hat{\mathbf{u}}$ is along $\mathbf{F}_{in}$:

$$\mathbf{a}_{rel}(t^{*+}) = (\mu - \mu_k)\,g\,\frac{\mathbf{F}_{in}}{|\mathbf{F}_{in}|}, \qquad |\mathbf{a}_{rel}| = (0.4 - 0.3)\cdot 9.81 = 0.981\ \text{m s}^{-2}$$

Components: $0.981\cos\psi = 0.9807$ m s^{-2} outward and $-0.981\sin\psi = -0.0250$ m s^{-2} along $\mathbf{e}_\varphi$: the block **lags behind** the table as it drifts out. It does not start along the radius.

Step 3. Cross-check in S_0

In S_0, $\mathbf{a} = \mathbf{F}_k/m = \mu_k g\,\hat{\mathbf{F}}_{needed}$ (inward and forward, 2.943 m s^{-2}), while the point of the table under the block has $\mathbf{a}_{drag} = \mu g\,\hat{\mathbf{F}}_{needed}$ (3.924 m s^{-2}, part (c)). With $\mathbf{v}_{rel} = \mathbf{0}$ the Coriolis term vanishes and $\mathbf{a}_{rel} = \mathbf{a} - \mathbf{a}_{drag} = (\mu_k - \mu)g\,\hat{\mathbf{F}}_{needed}$: the same result. In S_0 the block moves at 1.400 m s^{-1} **tangentially**; it is not "flung out along the radius". It simply cannot be pulled inward hard enough to stay on its circle, so its path opens out.

Step 4. Equal coefficients

If $\mu_k = \mu$, $\mathbf{a}_{rel}(t^{*+}) = \mathbf{0}$: the sliding starts smoothly, driven by the growth of Ω after t^*, but in the same direction.

KEY RESULTS

- In S_1: centrifugal + Euler $= -\mathbf{F}_{needed}$, of modulus μmg, pointing **outward and backward** (at ψ behind the outward radius).
- $\mathbf{a}_{rel}(t^{*+}) = (\mu - \mu_k)\,g$ in that direction: 0.981 m s^{-2}, components 0.9807 m s^{-2} along $\mathbf{e}_r$ and -0.0250 m s^{-2} along $\mathbf{e}_\varphi$.
- In S_0 the velocity is tangential (1.400 m s^{-1}, no radial launch) and the acceleration $\mu_k g = 2.943$ m s^{-2} is smaller than the $\mu g = 3.924$ m s^{-2} needed to follow the table.

Interpretation and checks

- **Dimensional check:** $\mu g/r_0$ and β are s^{-2}, so $\Omega^{*4} = (\mu g/r_0)^2 - \beta^2$ is s^{-4} and $t^* = \Omega^*/\beta$ is s; $r_0\Omega^2$, $r_0\beta$, μg are $\mathrm{m\,s}^{-2}$.
- **Limits:** $\beta \to 0$ (very slow spin-up) gives $r_0\Omega^{*2} = \mu g$, the classic result for uniform rotation, with $\psi \to 0$ and sliding along the outward radius in S_1; $\beta \to \mu g/r_0$ gives $t^* \to 0$ and $\psi \to 90^\circ$: the block slides straight backward.
- **Common mistakes in the exam:** (1) writing $r_0\Omega^2 \le \mu g$ and forgetting the tangential friction $m r_0 \beta$; (2) forgetting the Euler force $-m\dot{\boldsymbol{\omega}} \times \mathbf{r}$ in S_1; (3) putting $|\mathbf{F}| = \mu N$ before slip (static friction is only bounded by μN); (4) claiming the block flies off radially in S_0.